

Indexed pressure patches for efficient rigid-body contact

Viktor Jonsson's collision-model research Research checkpoint

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Outcome. A small pressure-patch candidate generates both force and independent torque from the same local deformation/friction calculation. An exact moment reduction speeds every tested synthetic indexed CPU batch. These are mechanical and numerical results: no new improvement against experimental measurements is claimed, and production collision examples remain unqualified.

What changed. Previously the new finite-patch branch handled pure axial spin only. The candidate now admits simultaneous sliding, axial spin, transverse rotation, asymmetric pressure, irregular footprints and local normal unloading. Bodies stay rigid; contact compression has three affine coordinates and the footprint has a small site/template representation. This is a frozen planar foundation reference, not full hydroelastic geometry or a deformation mesh.

Mechanics with the established notation

Keep angular momentum $\mathbf{L}$, numerical reference length ℓ , inertia $\mathbb{I}$, $\mathbf{V} = [\mathbf{v}; \ell\boldsymbol{\omega}]$ and $\mathbf{P} = [\mathbf{p}; \mathbf{L}/\ell]$. Contact impulses remain $\delta\mathbf{p}, \delta\mathbf{L}$; combined changes use uppercase $\Delta\mathbf{P}$. Each body retains only index k . In the body update, the contact-origin couple is additional to the force's center-to-contact moment:

$$\Delta\mathbf{L}_k = \mathbf{r}_k \wedge \delta\mathbf{p} + \delta\mathbf{L}.$$

Here the displayed contact acts with positive incidence sign; signed incidence supplies the opposite body's contribution. The article's transpose-only wedge reversal remains unchanged.

For one pressure site, $\mathbf{r} = [x, y, 0]^T$, $\mathbf{b} = [1, y, -x]^T$, $\mathbf{d}$ contains virtual compression/tilts and $\mathbf{q} = [v_z, \omega_x, \omega_y]^T$. Let $d = \mathbf{b}^T \mathbf{d}$, $u = \mathbf{b}^T \mathbf{q}$. Then $\dot{d} = -u$. With total stiffness K , damping C and positive site weight w , the unilateral normal contribution is

$$f = w \max(0, K \max(d, 0) - Cu) \quad (d > 0), \quad f = 0 \quad (d \leq 0).$$

Local tangential slip is $[v_x - \omega_z y, v_y + \omega_z x]^T$; its kinetic traction opposes this slip with magnitude $\mu_d f$. At zero local slip this kinetic branch is zero; static friction remains an unresolved additional branch. Summing $\mathbf{f}_j$ and $\mathbf{r}_j \wedge \mathbf{f}_j$ over site j produces resultant force and the independent contact couple. The same local friction budget supplies both.

Stored energy, release and an exact small matrix

Stored normal energy is $U = \frac{1}{2}K \sum_j w_j \max(d_j, 0)^2$. Normal dissipative power is $D_n = \sum_j (w_j K \max(d_j, 0) - f_j)u_j \geq 0$; kinetic sliding loss is $D_t = \mu_d \sum_j f_j |\mathbf{u}_j^{\text{slip}}| \geq 0$. The instantaneous work identity is

$$\mathbf{F} \cdot \mathbf{v} + \mathbf{T} \cdot \boldsymbol{\omega} + \dot{U} + D_n + D_t = 0.$$

Unloading/clipping keeps stored energy and its dissipation in the ledger instead of deleting a spring state. This isolated frozen transient does not implement state transport into a later moving footprint/contact.

Cache the small Gram matrix

$$\mathbb{G} = \sum_j w_j \mathbf{b}_j \mathbf{b}_j^T.$$

If all sites are compressed and trial pressures positive, the normal force/rolling-moment triple is exactly $\mathbb{G}(K\mathbf{d} - C\mathbf{q})$. If axial twist is zero, local tangential velocity is uniform and its force/moment also reduce using the same pressure moments. A conservative footprint-box test admits this branch. Other states retain site-level evaluation, including opening. This is an algebraic reduction, with no altered material constants or tolerances.

For circular Hertz weights, uniform compression and zero center slip, $T_x = -Ca^2\omega_x/5$ produces normal-deformation rolling resistance. Pure axial sliding gives $T_z = -3\pi\mu_d Na/16$ times spin sign, using the existing dynamic coefficient and physical radius a . This is not a new fitted coefficient, a universal material law or a new mechanics identity. Mixed states cannot simply apply both their separate maximum sliding force and maximum axial moment.

Input	Treatment
e_n, e_t	Existing restitution inputs kept separate; not double-imposed on top of compliant dynamics. Mapping to measured compliant response still required.
μ_s, μ_d	Fixed documented values remain fixed; this candidate uses only the dynamic branch.
μ_r	Retain documented rolling resistance with its physical length scale; do not add it again to pressure-induced losses.
K, C, a	Independent effective contact characterization/geometry. Current controls use explicit synthetic values, not rubber or rock measurements.
ℓ	Coordinate length, never a material/footprint radius.

Indexed computation and numerical verification

Use separate body-component arrays, flat contact fields and shared immutable footprint templates. A contact owns a range of signed incidences; each incidence has one owner body k , sign and lever arm. Site j owns one pressure sample. Gather contact motion, compute a local wrench, then scatter force and lever moment plus independent couple. No dense body-by-contact tensor or global inverse is needed for this local operation. Full rapid groups still require a coupled solve.

Vektor Flow’s Section 0 requires explicit reductions: for a contact-local view, $\mathbf{f}_x: \text{sum}_j(\mathbf{f}_{x_j})$. Repeated dimension names are not implicit sums. Keep one index per entity and ordered reductions where floating-point order is part of the contract. Current Python/C++ does not certify a VKF compiler port, ragged GPU gather/scatter, collision integrator or evolving-history semantics.

240 controls cover planar/spatial mechanics, 50 loaded and 190 partly open or clipped states, arbitrary world frames and body incidences. Maximum scaled instantaneous energy-rate residual is $1.92\text{e-}15$; indexed work/global momentum residual is $4.91\text{e-}16$. Analytic Hertz spin moment and cached normal Gram calculations agree near roundoff. The isolated normal transient tracks kinetic, stored and lost energy with maximum absolute residual $1.34\text{e-}09$ J against initial 0.5 J.

Directional compatibility remains a real issue. The user’s directions are full relative velocity $\hat{\mathbf{t}}$ and full relative angular velocity $\hat{\mathbf{s}}$, with

$$\delta\mathbf{p} = \delta p_n \hat{\mathbf{n}} + \delta p_t \hat{\mathbf{t}}, \quad \delta\mathbf{L} = \delta L_s \hat{\mathbf{s}} + \delta L_n \hat{\mathbf{n}}.$$

We do not redefine these directions. In one synthetic mixed state, orthogonal residuals outside those allowed spans are:

Footprint	Force residual (N)	Couple residual (N m)
hertz	0.442518	0.010372
ellipse	0.380684	0.005927
irregular	0.631587	0.012675

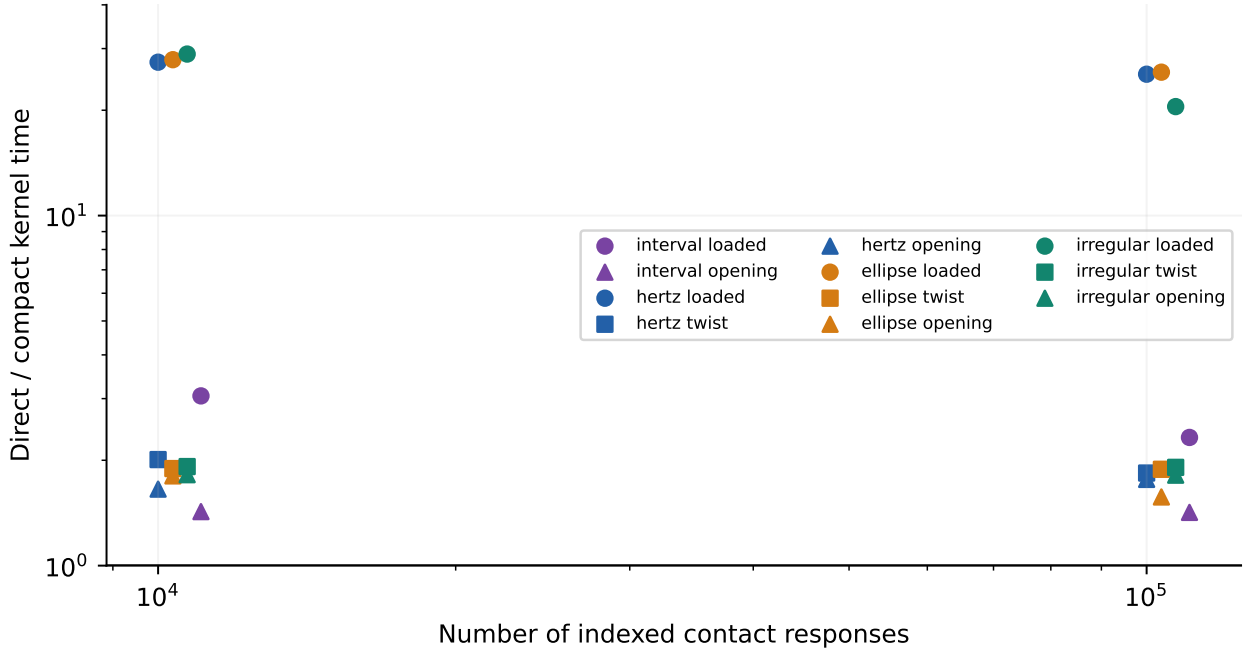
These residuals depend on the contact origin. Moving to the center of normal pressure removes the transverse couple; force residuals become 0.070009, 0.617335 and 0.192376 N respectively. The corresponding full relative velocity is evaluated at that physical point; $\mathbf{t}$ is not relabeled. 100 origin-shift and 300 length-scale controls preserve body torque, work and stored energy near roundoff. This does not prove every possible directional closure is incompatible. It shows that origin/pressure geometry and the remaining mixed-force discrepancy need reconciliation before equivalence/adoption. Zero velocity/spin needs a static direction convention too.

Efficiency with identical prescribed sites

Warning-free C++17 -O3 -Wall -Wextra -Werror, no fast-math. The timed operation includes body gather, patch force/couple, ordered body scatter and output resets. All 44 first-run batches (100, 10,000, 100,000 or one million responses) improve. A second 22-batch run alternates timing order and improves every batch too. 48 native/Python controls pass each run, maximum scaled error 2.28×10^{-15} . These are frozen responses, not interacting scenes.

The first-run million-response totals are below. The last column gives the alternating-order confirmation range for 10,000/100,000 responses, not a statistical confidence interval or a million-body result.

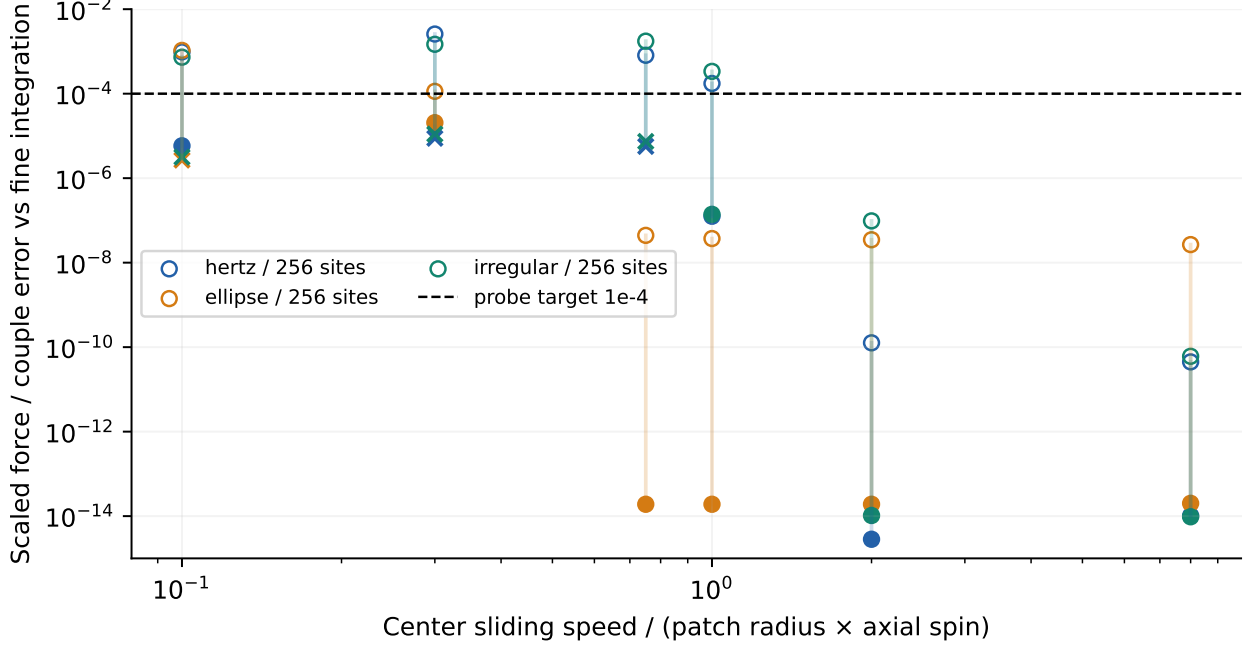
Case	Sites	Direct ms	Compact ms	Gain	Confirmation
interval / loaded	8	165.46	62.61	2.64	2.32–3.05
interval / opening	8	112.78	80.02	1.41	1.42–1.43
hertz / loaded	128	1291.78	51.87	24.90	25.35–27.43
hertz / mixed twist	128	1295.58	681.93	1.90	1.84–2.01
hertz / opening	128	1226.39	673.86	1.82	1.65–1.76
ellipse / loaded	128	1291.44	52.00	24.84	25.69–27.89
ellipse / mixed twist	128	1291.59	680.69	1.90	1.88–1.89
ellipse / opening	128	1224.99	672.18	1.82	1.57–1.80
irregular / loaded	128	1279.16	51.36	24.91	20.48–28.94
irregular / mixed twist	128	1278.68	675.55	1.89	1.91–1.92
irregular / opening	128	1223.03	668.34	1.83	1.81–1.82



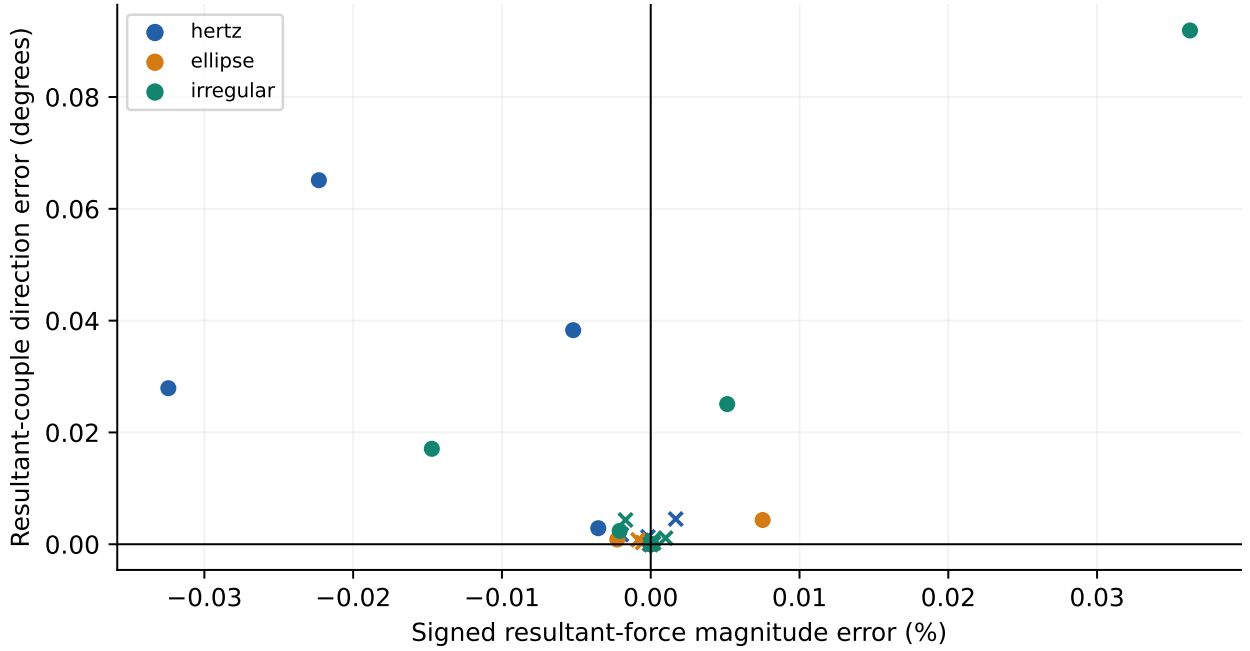
The large no-twist gains come from eliminating site loops. Mixed-spin/opening gains come from compact component arithmetic with the same sites. Footprint generation, frame transformations, shear/static branches, time integration, detection and group solves are outside timing. Allocations are excluded. First run's inaccurate array-size estimate is retained and corrected in v2; timings and wrenches are unchanged. Host load affects timing, and no end-to-end claim or renewed 2x gate is made.

Patch accuracy: difficult states stay visible

The fast exact reduction has the same prescribed-site output; that alone cannot certify a continuous pressure-patch approximation. Near a local slip-zero, fixed coarse quadrature is inaccurate and convergence can be nonmonotone. We probe 18 mixed states for circular/elliptic/irregular shapes against a 147,456-site reference. Two consecutive small changes are required to admit a bounded refinement estimate. 12/18 estimates admit; all 18 finest returned values happen to satisfy the 10^{-4} scaled-error target, maximum 2.06e-05. The six declined estimates remain declines. This finite check is not a rigorous bound for arbitrary states, and 65,536-site worst branches are too expensive for a cheap general contact law.



Open dots: fixed 256 sites. Filled dots: admitted refinement estimates. Crosses: budget-ended declines. Vertical segments connect the same inputs. Error is the maximum of force error/ N and couple error/ $(N a)$. It compares numerical integration of the candidate, not reality.



Signed force-size error and couple-angle error against the same fine-patch input, with shape colors. Dots use 256 sites; crosses use 4,096. The reference point is the origin. These are numerical error points, not experimental accuracy points.

Where the real-data comparison stands

This checkpoint adds no measured collision records and fits no coefficients. The earlier comparison tables remain unchanged:

Dataset	Status retained from the experimental report
Glass, 24 rows	Fixed documented comparator: normal/tangent COM RMSE 0.036674/0.017805 m/s; spin reconstructed and nonindependent. Not validation of this patch.
Ball/surface, 8 summaries	Target-supplied tangential restitution includes measured spin; conditional comparison. Earlier shared moment fit worsens error and remains rejected.
Rocks, 75 rows	Historical fitted sphere proxy: 25-height-test RMSE 1.004366/1.304693 m/s and 12.876456 rad/s. Missing actual attitude/inertia/footprints; damping-only extension worsens response and remains rejected.
Rolling, 17 points	Earlier relaxation reduction improves 8 reused evaluation points by 51.53/67.46/33.71 percent across three specimens. Exploratory reuse, not new blind validation or identified bulk relaxation.

Next authenticity work. Recover matched footprint/pressure or force-time measurements and independently characterized stiffness/damping/shear response; keep documented friction and restitution fixed. A finite shear-history/mode branch is needed for grip/slip reversal and restitution, and local yield/indentation geometry for damaged concrete. Reconcile the admissible directional map before adoption. Replace expensive singular mixed-state quadrature with a reduced, energy-consistent approximation that passes the same force/couple error controls. No per-case endpoint correction or newly invented material coefficient is added.

Precedent. Elandt, Drumwright, Sherman and Ruina (IROS 2019), [pressure-field contact](#), motivates resultant force/moment from locally compliant traction while retaining rigid bodies. Our flat foundation is simpler and does not implement their intersection geometry. [Drake's guide](#) describes the pressure-patch approach and warns that automatic/default material values need application-specific characterization. No novelty or measured-material authenticity follows from adopting a patch representation.

Reproduce. See README and INDEXED-DESIGN. Authoritative new evidence: audit-v2, refinement-v2, benchmark-v1 full ladder and benchmark-v2 alternating-order confirmation. v1 metadata-label mistakes remain retained with explicit correction receipt and unchanged numerical values. No production source is changed.